

Final Exam: Chapter 6 and 7

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1 Introduction

2 Chapter 7: Terms of Trade

- Terms of Trade and Trade Balance
- Terms of Trade and Business Cycle
- The MX Model
- Extension: Alternative Detrending Methods [New]

3 References

Motivation

Why does fiscal targeting matter?

Conditional Cash Transfers (CCT):

- Important policy instrument, widely used to **redistribution**.
- Standard models assume **Ricardian Equivalence** and representative-agent framework (**gali2015monetary**).
- Households highly differ in **Marginal Propensity to Consume (MPC)**, wealth, labor status, etc (**kaplan2014model**; **kaplan2018monetary**).

Why HANK:

- Endogenous MPC heterogeneity and liquidity-constraint **amplify aggregate effects**. Inequality matters for macro (**auclert2018inequality**).
- Fiscal policy highly depends on **who receives the transfer**, and on the financing method (**auclert2023trickling**).

What are the aggregate and distributional effects of CCT as a stabilization tool?

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Chapter 7 – Replication and Extension

- We **replicate the empirical analysis** in Chapter 7.
- The replication is conducted in Python and reproduces the results reported in the book.
- We then extend the analysis to assess the robustness of the results to alternative detrending methods.

Extension: Does the detrending method matter?

We compare the authors' **quadratic detrending** with:

- Hodrick–Prescott (HP) filter;
 - Hamilton filter.
-
- This extension is motivated by Carlos Eduardo's comment that the choice of detrending method may affect the results.

Motivation: Why Terms of Trade Matter?

- The **Terms of Trade (ToT)** capture the relative price of a country's exports and imports:

$$ToT_t = \frac{P_t^X}{P_t^M}.$$

- For emerging economies, ToT fluctuations can generate large changes in **national income and purchasing power**.
- A deterioration in the ToT reduces the amount of imports that can be purchased with a given volume of exports.
- As a result, ToT shocks can propagate to key macroeconomic variables such as **output, consumption, investment, and the trade balance**.

Key Motivation

How important are Terms-of-Trade shocks for macroeconomic fluctuations?

From Terms of Trade to the MX Model

1 Terms of Trade

► Terms of Trade and the Trade Balance

- We study how ToT shocks affect the **trade balance**.
- We estimate the persistence of ToT fluctuations and analyze the dynamic response of the trade balance.
- We also test whether the response depends on the **persistence of ToT shocks**.

► Terms of Trade and Business Cycles

- We extend the analysis to **output, consumption, investment, and the trade balance**.
- We quantify how these variables respond to ToT shocks.
- We estimate the share of macroeconomic fluctuations **explained by Terms-of-Trade shocks**.

2 MX Model

- A **small open-economy model with incomplete markets** used to quantify the macroeconomic effects of ToT shocks.
- We use the estimated ToT process to **parameterize the model**.
- We evaluate how ToT shocks propagate to **output, consumption, investment, and the trade balance**.

3 Robustness to Detrending [Extra Work]

- We test whether the results are sensitive to the **detrending method**, comparing quadratic detrending with the HP and Hamilton filters.

Chapter 7: Terms of Trade

Terms of Trade and Trade Balance

Terms of Trade and the Trade Balance

Empirical Question

Does an improvement in the Terms of Trade cause an **improvement or deterioration of the trade balance**?

- We estimate a country-specific SVAR for the **Terms of Trade** and the **trade balance**. More on Terms of Trade
- The cyclical components are obtained using **quadratic detrending**.¹

$$\begin{aligned}\widehat{tot}_t &= \rho \widehat{tot}_{t-1} + \pi \varepsilon_t^1, \\ \widehat{tb}_t &= \alpha_0 \widehat{tot}_t + \alpha_1 \widehat{tot}_{t-1} + \rho_2 \widehat{tb}_{t-1} + \sqrt{\sigma_{22}} \varepsilon_t^2.\end{aligned}$$

- We identify ε_t^1 as the **Terms-of-Trade shock**:
 - ▶ ToT is assumed to be contemporaneously unaffected by other shocks;

$$\begin{bmatrix} \widehat{tot}_t \\ \widehat{tb}_t \end{bmatrix} = h_x \begin{bmatrix} \widehat{tot}_{t-1} \\ \widehat{tb}_{t-1} \end{bmatrix} + \Pi \begin{bmatrix} \varepsilon_t^1 \\ \varepsilon_t^2 \end{bmatrix}, \quad h_x \equiv \begin{bmatrix} \rho & 0 \\ \alpha_0 \rho + \alpha_1 & \rho_2 \end{bmatrix}, \quad \Pi \equiv \begin{bmatrix} \pi & 0 \\ \alpha_0 \pi & \sqrt{\sigma_{22}} \end{bmatrix}.$$

¹Later, we assess robustness using alternative detrending methods.

How Should the Trade Balance Respond to ToT Shocks?

- A positive ToT shock may improve the trade balance, but the magnitude of the response depends on the underlying mechanism.

1. Harberger–Laursen–Metzler (HLM)

A ToT improvement raises real income and savings, improving the trade balance.

$$\frac{\partial tb_t}{\partial tot_t} = \frac{1 - \alpha}{1 + \mu - \alpha} q > 0$$

Prediction: a positive ToT shock improves the trade balance.

Derivation

2. Obstfeld–Svensson–Razin (ORS)

The response depends on the **persistence** of the ToT shock:

$$\frac{\partial tb_t}{\partial tot_t} = \frac{1 - \rho}{1 + r - \rho} \quad \rho \uparrow \Rightarrow \Delta tb \downarrow$$

Prediction: more persistent shocks generate a smaller improvement.

Derivation

Testing the ORS Effect

Testable Prediction

The ORS effect predicts that countries with more persistent ToT shocks should exhibit a **smaller trade-balance response**.

From the country-specific SVAR, for each country i we recover:

$$\underbrace{\rho_i}_{\text{Persistence of ToT shocks}} \quad \text{and} \quad \underbrace{\frac{\pi_{21,i}}{\pi_{11,i}} \times 10}_{\text{TB impact response to a 10\% ToT shock}}.$$

The ORS hypothesis therefore implies:

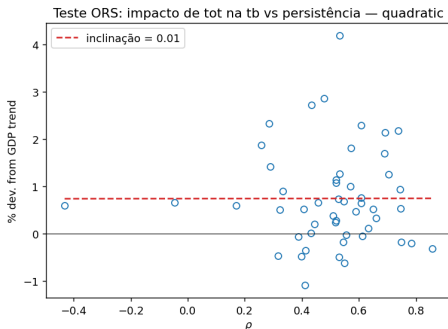
$$\boxed{\rho_i \uparrow \implies \frac{\pi_{21,i}}{\pi_{11,i}} \times 10 \downarrow}$$

Across countries, we should observe a negative relationship.

Testing the ORS Effect: Cross-Country Evidence

- We compare the estimated ToT persistence, ρ_i , with the trade-balance impact response across countries.
- Under ORS, the fitted relationship should be **negative**.

Additional ORS Tests



Result

We find **no clear negative relationship** between ToT persistence and the trade-balance impact response.

Chapter 7: Terms of Trade

Terms of Trade and Business Cycle

Terms of Trade as Drivers of Business Cycles

A Broader Question

How important are Terms-of-Trade shocks as **drivers of business cycles**?

- We extend the previous empirical analysis from **2 to 5 variables**.

$$\begin{aligned}x_t^1 &= \rho x_{t-1}^1 + u_t^1, \\x_t^2 &= \alpha_0 x_t^1 + \alpha_1 x_{t-1}^1 + \rho_2 x_{t-1}^2 + u_t^2,\end{aligned}$$

where

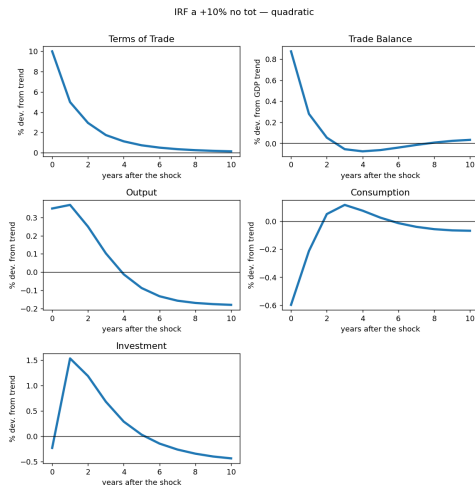
$$x_t^1 = \widehat{tot}_t, \quad x_t^2 = \begin{bmatrix} \widehat{tb}_t \\ \widehat{y}_t \\ \widehat{c}_t \\ \widehat{i}_t \end{bmatrix}.$$

- As before, variables are detrended using a **quadratic trend**.
- ToT shocks are identified as the only structural innovation affecting ToT contemporaneously.

Macroeconomic Responses to a ToT Shock

Following a **10% improvement in the ToT**:

- **Trade balance improves**
Higher purchasing power initially raises external saving.
- **Output initially expands**
Higher real income stimulates economic activity.
- **Consumption falls on impact**
Part of the initial income windfall is saved; consumption then recovers.
- **Investment initially falls, then expands**
The response is delayed, peaking one year after the shock.



A positive ToT shock generates an initial **income and purchasing-power gain**, which propagates through saving, consumption, output, and investment.

How Important Are ToT Shocks for Business Cycles?

- Impulse responses show **how** the economy reacts to ToT shocks. To assess their quantitative importance, we compute the fraction of each variable's variance attributable to ToT shocks:

$$\text{Variance Share}_j = \frac{\text{Var}(x_{j,t} \mid \varepsilon_t^{\text{ToT}})}{\text{Var}(x_{j,t})} \times 100.$$

	ToT	Trade Balance	Output	Consumption	Investment
Cross-Country Mean	100.0	17.6	11.6	13.5	12.5
Cross-Country Median	100.0	12.2	9.8	10.6	10.2
Median Absolute Deviation	0.0	11.4	8.8	9.1	8.3
Using Mean of h_x and Π	100.0	7.0	2.9	1.5	2.2
Panel Estimation	100.0	4.0	1.3	1.2	0.5

- Values may differ slightly from those reported in the book because we use the **non-rounded estimates** in our calculations.
- Using the cross-country median, ToT shocks explain about **10–12%** of fluctuations in the main macroeconomic aggregates.

Main Result

Terms-of-Trade shocks are a **relevant but limited** source of business-cycle fluctuations in emerging and poor economies.

Chapter 7: Terms of Trade

The MX Model

The MX Model: Why Do We Need It?

- So far, the empirical SVAR has shown how the economy **responds to Terms-of-Trade shocks**.
- But the previous one-good framework treats the ToT shock essentially as a productivity shock.
- This implies an unrealistic economy in which:
 - ▶ virtually all domestic production is exportable;
 - ▶ virtually all domestic absorption is imported.
- In the data, however, countries both **produce and consume** importable and exportable goods.

Key Innovation of the MX Model

The economy explicitly contains two sectors: **Importables** (m) and **Exportables** (x).

Empirical SVAR $\implies$ Observed responses

MX Model $\implies$ Structural mechanism

The MX Model: Households

Households choose consumption, sector-specific labor, investment, capital accumulation, and external debt:

$$\max E_0 \sum_{t=0}^{\infty} \beta^t U(c_t, h_t^m, h_t^x)$$

subject to

$$\begin{aligned} c_t + i_t^m + i_t^x + \Phi_m(k_{t+1}^m - k_t^m) + \Phi_x(k_{t+1}^x - k_t^x) + d_t \\ = \frac{d_{t+1}}{1+r_t} + w_t^m h_t^m + w_t^x h_t^x + u_t^m k_t^m + u_t^x k_t^x. \end{aligned}$$

Capital evolves separately in each sector:

$$k_{t+1}^j = (1 - \delta)k_t^j + i_t^j, \quad j \in \{m, x\}.$$

- Labor and capital are **sector-specific**.
- Adjustment costs prevent instantaneous reallocation across sectors.
- Households can borrow or lend internationally.

The MX Model: Production Structure

Sectoral production

$$y_t^m = F^m(k_t^m, h_t^m), \quad y_t^x = F^x(k_t^x, h_t^x).$$

Firms maximize profits:

$$p_t^j F^j(k_t^j, h_t^j) - w_t^j h_t^j - u_t^j k_t^j,$$

implying

$$p_t^j F_1^j = u_t^j, \quad p_t^j F_2^j = w_t^j.$$

Final-good production

$$A(a_t^m, a_t^x)$$

combines importable and exportable goods using an **Armington aggregator**.

$$A_1(a_t^m, a_t^x) = p_t^m, \quad A_2(a_t^m, a_t^x) = p_t^x.$$

Market Clearing and the Terms-of-Trade Shock

Domestic final-good market clearing:

$$c_t + i_t^m + i_t^x + \Phi_m(\Delta k_t^m) + \Phi_x(\Delta k_t^x) = A(a_t^m, a_t^x).$$

Imports and exports are

$$m_t = p_t^m (a_t^m - y_t^m), \quad x_t = p_t^x (y_t^x - a_t^x).$$

External debt evolves according to

$$\frac{d_{t+1}}{1 + r_t} = d_t + m_t - x_t.$$

Terms of Trade

$$tot_t = \frac{p_t^x}{p_t^m}, \quad \widehat{tot}_t = \rho \widehat{tot}_{t-1} + \pi \varepsilon_t^{tot}.$$

- As in the SVAR, the economy is small and takes the ToT as **exogenous**.
- Importantly, the same estimated ρ and π from the empirical exercise discipline the model.

From the Model to the Data

- With multiple goods, model variables cannot be compared directly with the empirical aggregates used in the SVAR.
- The WDI measures real GDP using a **Paasche implicit price deflator**.
- Therefore, the model constructs theoretical counterparts consistent with the empirical measures.

Nominal GDP is

$$P_t^m y_t^m + P_t^x y_t^x.$$

Using the Paasche deflator, model-consistent real GDP becomes

$$GDP_t^{real} = p^m y_t^m + p^x y_t^x$$

where steady-state relative prices are used as base-period prices.

Why This Matters

The MX model and the empirical SVAR must measure **the same economic objects** before their predictions can be compared.

Functional Forms and Economic Frictions

Preferences: GHH

$$U(c, h^m, h^x) = \frac{[c - G(h^m, h^x)]^{1-\sigma} - 1}{1-\sigma}.$$

Sectoral labor

$$G(h^m, h^x) = \frac{(h^m)^{\omega_m}}{\omega_m} + \frac{(h^x)^{\omega_x}}{\omega_x}.$$

Production: Cobb-Douglas

$$F^j(k^j, h^j) = A^j (k^j)^{\alpha_j} (h^j)^{1-\alpha_j}.$$

Capital adjustment costs

$$\Phi_j(z) = \frac{\phi_j}{2} z^2.$$

Debt-elastic premium

$$p(d) = \bar{p} + \psi(e^{d-\bar{d}} - 1).$$

Armington aggregator

$$A(a^m, a^x) = \left[\chi (a^m)^{1-1/\mu} + (1-\chi) (a^x)^{1-1/\mu} \right]^{\mu}.$$

- ϕ_m, ϕ_x : slow capital reallocation.
- ω_m, ω_x : imperfect labor mobility.
- μ : substitution between importables and exportables.
- ψ : stabilizes external debt.

Parameterization of the MX Model

The model combines **calibration** and **country-level estimation**.

❶ Parameters inherited from standard SOE models

$$\sigma, \delta, r^*, \bar{p}, \alpha_m, \alpha_x, \omega_m, \omega_x.$$

❷ Parameters disciplined by sectoral trade data

$$\mu, \chi, \bar{d}, A_x.$$

They match average trade-balance, export, and exportable-output shares.

❸ Country-specific parameters

$$\rho, \pi, \phi_m, \phi_x, \psi.$$

Here, ρ and π come directly from the **estimated ToT process**.

	ρ	π	ϕ_m	ϕ_x	ψ
Median	0.53	0.09	1.82	1.56	0.18

Connection to the Empirical Exercise

The estimated ToT process from the first activity **feeds directly into the structural MX model**.

What Does the MX Model Predict?

Following a positive ToT shock:

- production shifts from **importables to exportables**;
- imports rise as domestic demand for importables increases;
- exports rise as exportable production expands;
- GDP and consumption increase;
- the trade balance **deteriorates on impact**.

Key Tension

The model fails to reproduce the empirical **HLM effect**: in the SVAR, the trade balance improves on impact.

	MX Model	Empirical SVAR
Trade Balance	27%	12%
Output	18%	10%

Chapter 7: Terms of Trade

Extension: Alternative Detrending Methods [New]

Motivation: Does the Detrending Method Matter?

- To implement this extension, we had to return to the **raw data** and reconstruct the empirical analysis from the beginning.
- This made the replication more demanding, but also allowed us to reproduce the analysis under **three alternative specifications**:

Quadratic Trend (Baseline)

HP Filter

Hamilton Filter

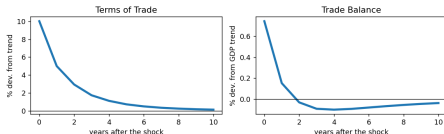
What do we want to know?

- 1 Does the relationship between **ToT shocks and the trade balance** change across detrending methods?
- 2 Does the **ORS test** remain inconclusive?
- 3 Do **macroeconomic variables** respond differently to ToT shocks?
- 4 Do ToT shocks explain **more or less of business-cycle fluctuations**?

Robustness: Does Detrending Matter?

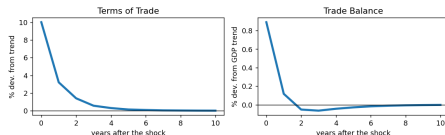
Quadratic (Baseline)

IRF a +10% no tot — quadratic



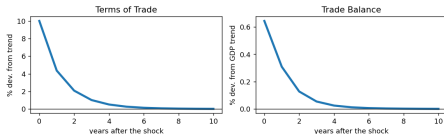
HP Filter

IRF a +10% no tot — hp



Hamilton

IRF a +10% no tot — hamilton



What Changes?

HP Filter

- Very similar to the baseline.
- Same short-lived improvement in the trade balance.

Hamilton Filter

- The response becomes more persistent.
- No short-run trade-balance reversal.
- ToT shocks are approximately **2× more volatile**.

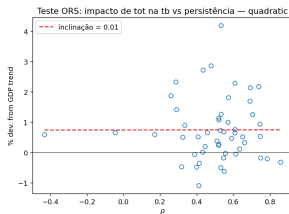
Robustness: Does the ORS Result Depend on Detrending?

ORS Prediction

More persistent ToT shocks should generate a **smaller** trade-balance response:

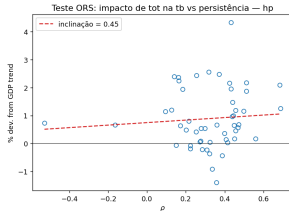
$$\rho_i \uparrow \implies IR_i^{TB}(0) \downarrow.$$

Quadratic



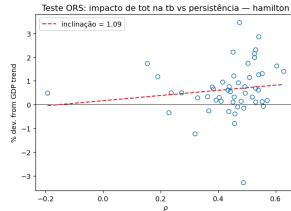
$$\hat{\beta} = 0.01$$

HP Filter



$$\hat{\beta} = 0.45$$

Hamilton



$$\hat{\beta} = 1.09$$

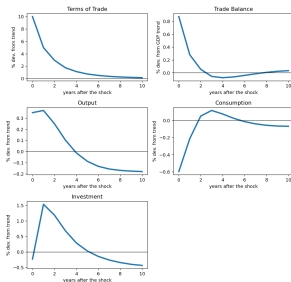
Main Result

The ORS prediction is **not recovered under any detrending method**. All estimated slopes are non-negative, and the relationship becomes more positive under HP and Hamilton detrending.

Robustness: Macroeconomic Responses Across Detrending Methods

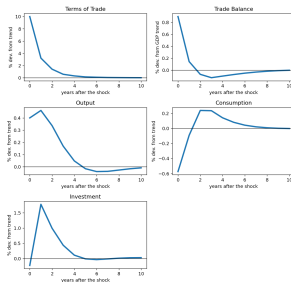
Quadratic (Baseline)

IRF a = 10% no tot — quadratic



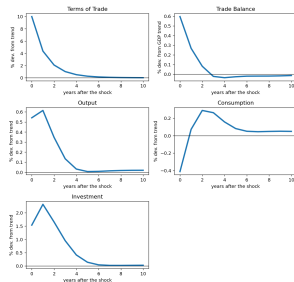
HP Filter

IRF a = 10% no tot — hp



Hamilton Filter

IRF a = 10% no tot — hamilton



Takeaway

The main qualitative results are **robust to the HP filter**. The **Hamilton filter**, however, produces noticeably stronger macroeconomic responses, consistent with the larger volatility of the identified ToT shocks.

Robustness: How Much Do ToT Shocks Explain?

- We recompute the variance contribution of ToT shocks under each detrending method.
- We focus on the **cross-country median** to compare the quantitative importance of ToT shocks.

	Trade Balance	Output	Consumption	Investment
Quadratic (Baseline)	12.2	9.8	10.6	10.2
HP Filter	13.2	12.9	14.7	14.1
Hamilton Filter	13.6	11.3	11.3	10.5

- **Trade Balance:** remarkably stable across methods, at around 12–14%.
- **HP Filter:** increases the contribution of ToT shocks, especially for consumption and investment.
- **Hamilton Filter:** remains relatively close to the quadratic baseline in terms of variance explained.

Takeaway

The conclusion is broadly robust: ToT shocks explain a **meaningful but limited share** of business-cycle fluctuations, although their quantitative importance is somewhat larger under the HP filter.

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Thank You!

Questions? Comments? Suggestions?

Appendix: Harberger–Laursen–Metzler Effect

Environment

A static small open economy with Keynesian consumption and import demand:

$$c_t = \bar{c} + \alpha y_t, \quad m_t = \mu y_t, \quad x_t = q \text{ tot}_t,$$

where $0 < \alpha < 1$, $\mu > 0$, and q is the volume of exports.

Goods-market equilibrium

$$y_t = c_t + g + i + x_t - m_t \implies y_t = \frac{\bar{c} + g + i + q \text{ tot}_t}{1 + \mu - \alpha}.$$

Using $tb_t = x_t - m_t$, the equilibrium trade balance is

$$tb_t = \frac{1 - \alpha}{1 + \mu - \alpha} q \text{ tot}_t - \frac{\mu(\bar{c} + g + i)}{1 + \mu - \alpha}$$

and therefore

$$\frac{\partial tb_t}{\partial \text{tot}_t} = \frac{1 - \alpha}{1 + \mu - \alpha} q > 0$$

Appendix: Obstfeld–Svensson–Razin Effect

Environment

A small open endowment economy in which households smooth consumption:

$$\max_{\{c_t, d_t\}} E_0 \sum_{t=0}^{\infty} \beta^t \left[-\frac{1}{2} (c_t - \bar{c})^2 \right]$$

subject to

$$d_t = (1 + r)d_{t-1} + c_t - tot_t, \quad tot_t = \rho tot_{t-1} + \varepsilon_t^{tot}.$$

Consumption smoothing implies that the trade balance responds according to

$$tb_t = rd_{t-1} + \underbrace{\frac{1 - \rho}{1 + r - \rho}}_{\text{response to ToT}} tot_t$$

Hence,

$$\frac{\partial tb_t}{\partial tot_t} = \frac{1 - \rho}{1 + r - \rho} > 0,$$

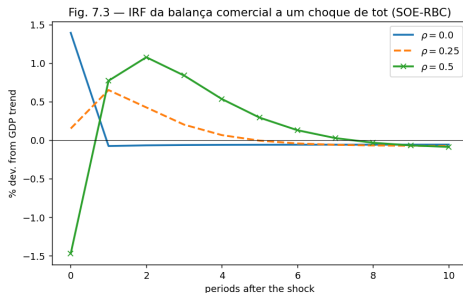
$$\frac{\partial^2 tb_t}{\partial tot_t \partial \rho} = -\frac{r}{(1 + r - \rho)^2} < 0$$

Appendix: ORS – Persistence and Trade-Balance Response

ORS Prediction

More persistent ToT shocks should generate a **smaller trade-balance response**:

$$\rho \uparrow \implies IR^{TB}(0) \downarrow$$



Result

The model illustrates how the trade-balance response changes with the persistence of the ToT shock.

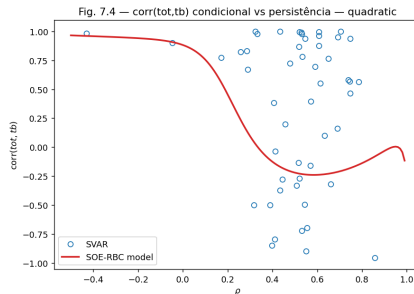
Note: Results using the HP and Hamilton filters are also available in the replication folder.

Appendix: ORS – Conditional Correlations

Alternative ORS Prediction

The SOE–RBC model links ToT persistence to the conditional correlation between ToT and the trade balance:

$$\rho \uparrow \implies \text{corr}(tot, tb \mid \varepsilon^{tot}) \downarrow$$



Result

The SVAR estimates show **no systematic relationship** matching the model prediction.

Note: Results using the HP and Hamilton filters are also available in the replication folder.